

NAPLPS: A New Standard for Text and Graphics

Part 4: More Advanced Features and Conclusions

*A standard way to encode color mapping and animation,
closing with some predictions on how NAPLPS
will be used by personal computers.*

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This is the fourth and final part of this series of articles on NAPLPS, the North American Presentation-Level-Protocol Syntax, a new communications standard for encoding both textual and graphics information. With this standard, graphics information can be sent from computer to computer, or from peripheral to peripheral, with little regard to inherent differences in the display capabilities of the various machines available. Such a standard will be particularly important for proposed mass-market data-communications systems such as videotex.

The first part of the series presented an overview of NAPLPS. The second part gave a detailed analysis of its basic features. And last month, some of the advanced features of NAPLPS were discussed. This month I will continue that discussion and conclude with some predictions about the future of NAPLPS.

About the Author

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These predictions are based on the premise that personal computing has *not* yet reached the majority of people in the world. For that to happen, personal computers must be very easy to use. Those readers who may be dreaming about the day when everyone will be assembling S-100 kits and toggling in bootstrap pro-

grams should probably reassess their thinking.

Advanced Color Capabilities

As was mentioned in part 1, NAPLPS supports a variety of color modes. The most primitive one (mode 0) is quite portable and will be used in the majority of applications. As was previously described, colors are specified in color mode 0 by indicating the relative amounts of red, green, and blue that should be mixed to form the desired color.

Color modes 1 and 2, however, use a technique called *color maps* or *color tables*. As shown in figure 1, a color table is a set of indexes and their corresponding colors. These are tied together using the Set Color and Select Color commands. As can be seen in the figure, a given mixture of primary colors can be used more than once, and all the indexes do not have to be used. Note that the index value does *not* have any direct relationship with a hardware *drawing value* at this point.

In order to make an entry in the color table, you must first use the Select Color command to specify a

INDEX	COLOR		
	RED	GREEN	BLUE
0	0.2	0.3	0.0
4	0.0	0.0	0.0
5	0.5	0.8	0.8
6	0.0	0.8	0.8
25	1.0	1.0	1.0
26	0.5	0.5	0.5
27	0.6	0.0	1.0
31	0.0	1.0	0.0

Figure 1: A typical color table used with color modes 1 and 2 in NAPLPS. Each index value corresponds to a specific color. Note that the index values are arbitrary and that no index value appears more than once. Each color is represented as a mixture of the primary colors red, green, and blue.

certain index (see figure 2). This index value is encoded in the byte or bytes following the command and usually ranges from 0 to 63, although indexes as high as 16,777,215 can be encoded. After an index is chosen, you can then use the Set Color command to specify the mixture of red, green, and blue values that should be associated with the index.

When you want to select a color for drawing in color modes 1 and 2, you must specify the index for that color—not the actual color. The primary difference between modes 1 and 2 is that mode 2 allows you to specify a background color for text characters.

As shown in figure 3, when a picture is drawn into display memory, a drawing value is allocated to each index. This value is written into the display memory. The color information currently associated with each index is put into a hardware register associated with each drawing value, and anything drawn with that display-memory value will have that color. Note that a display value is allocated to an index only when drawing occurs. Also, each display value is allocated to only one index.

In order to create blinking and animation effects, the color associated with an index can be changed using the Select/Set sequence de-

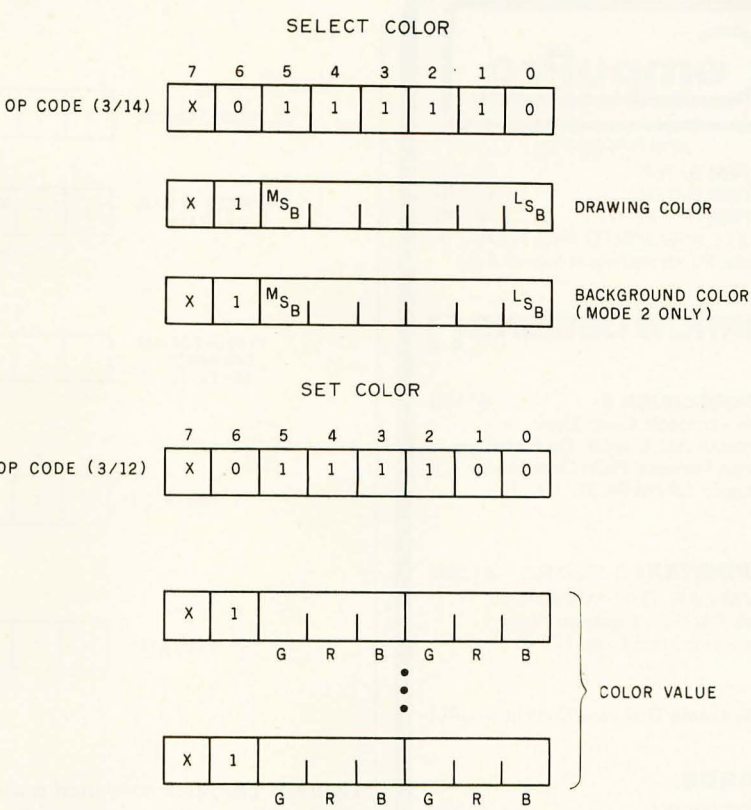


Figure 2: The Select Color and Set Color commands used to specify a color-table index value and a red, green, and blue mixture, respectively. (The codes 3/14 and 3/12 are an alternate way of indicating the hexadecimal numbers 3E and 3C.) In color mode 2, two indexes can be specified in the Select Color command if a background color for text is desired. The normal sequence used to load the color table is to first select an index and then set a color for that index. These actions cause a relationship to be established between an index value and a color. After a table is set up, each set color can be used by using the Select command. Also, the color associated with each index value can be changed by using another Set Color command.

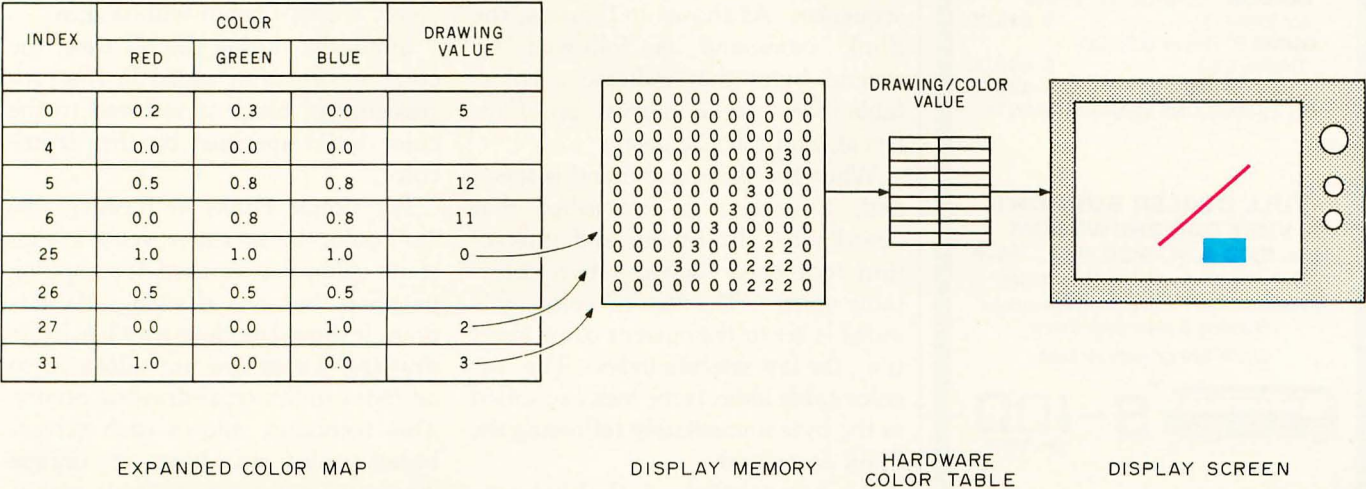


Figure 3: A typical color-table implementation. Note that special color-map hardware is required. The drawing value is the one used when drawing occurs in the display memory. It is an index into the hardware-based color map.

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BLINK

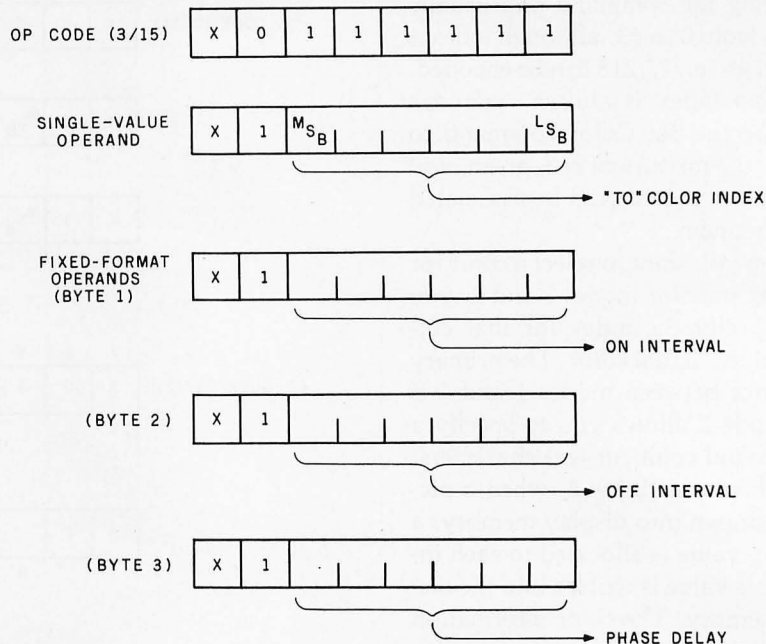


Figure 4: The Blink command is used to establish one or more asynchronous blink processes. These processes automatically modify the contents of the color map. In short, the Blink command causes the color of the current color index to be temporarily changed to that of the "to" color index. The on, off, and phase-delay times are specified in 1/10-second resolution, which allows this system to be compatible with both North American (60 frames per second) and European (50 frames per second) television systems. These times are used to control how often the color map is modified.

scribed above. This change will cause everything on the display screen that has been drawn with that value to change color.

Color-Table Animation

The Blink command is used to set up automatic color-table animation sequences. As shown in figure 4, the Blink command is followed by several bytes that indicate a color-table index, an on interval, an off interval, and a phase delay.

When the Blink command is specified, a process is established that coordinates the timing and interaction that occur between two color-table entries. The "from" color-table index is set to the current color index (i.e., the last selected index). The "to" color-table index is the index specified in the byte immediately following the Blink command.

As shown in figure 5, the blink processes are independent asynchronous activities that copy values from one entry in the color table to another.

During the on interval, the color corresponding to the index specified by the "from" color is saved in an area of memory called the process-control block. Then the color corresponding to the index specified by the "to" index is copied to the "from" index. If the new color is different than the old color, a visual result will be seen.

Similarly, during the off time, the color information saved in the process-control block is restored to the color index specified by the "from" color.

For simple blinks or flashing, the "to" color index can specify a constant color that is used for copying purposes but not drawing. At this point it should be clear why hardware drawing values are not allocated to an index until actual drawing occurs. This technique allows each screen-based color to have a unique "from/to" color pair without requiring that colors be shared.

If you are following carefully, you will note that, at the beginning of the

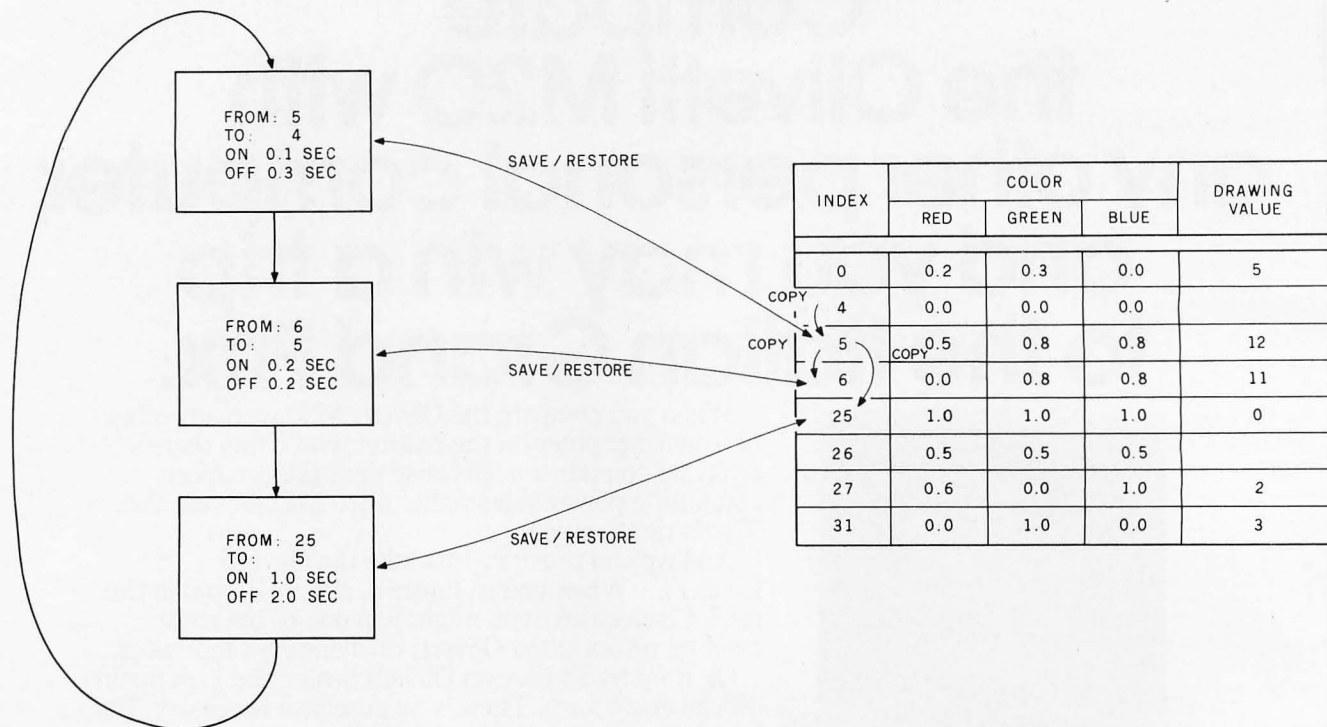


Figure 5: An example of three blink processes used together. The first process saves the contents of the color map at index 5 and then copies the contents of index 4 into index 5. After 0.1 second, the original contents of index 5 are restored for 0.3 second. The second process saves the contents of the map at index 6 and copies the contents of index 5 into index 6. Note that the second process can copy the contents of index 5 during the time it has been changed by process 1, causing a variety of effects. The third process, which runs very slowly, saves the contents of index 25 and also copies the contents of index 5 into index 25. When changes are made to the colors in indexes 5, 6, and 25, the corresponding hardware color-map registers 12, 11, and 0 are changed, causing visual changes to the screen.

on time, the color in the "to" color index is copied to the "from" color index. This copy is made independently of any blink activity that may be set for the "to" color. Thus, this copying could occur during a time when the "to" color has been changed by another blink process. The result is that multiple blink processes can be set up to allow colors to ripple around the color table in regular and irregular patterns. These patterns can be used to produce dramatic animation effects under full control of the terminal and without the need for host interaction.

These animation effects, plus the other features described in the previous articles, should establish NAPLPS as the most extensive information-exchange protocol available. As time goes on, it is expected that NAPLPS will replace ASCII (American National Standard Code for Information Interchange) as the standard for information inter-

change. As this occurs, almost every area of computing will be affected. In order to prepare for this impact, we need to look into the future to see how NAPLPS will help shape the world.

Predictions and Conclusions

If one sits back and attempts to predict the future of personal computing and information exchange, some obvious predictions come to mind:

1. Integrated text and graphics will be essential in all visual information exchange.
2. Personal computers must be designed to be so easy to operate that a casual user can begin doing useful things immediately.
3. A personal computer will be used as a *link* to the rest of the world rather than a *diversion* from it.
4. People's efficiency will be increased by allowing concurrent activities.

5. The average personal computer user will be more of a *consumer* than a *producer*.

It should be noted that I have said nothing about the size of memory chips or the density of disk drives. These predictions involve only functionality and the user's view of computing capability. I am a firm believer that the consumer will be the ultimate decision maker when the fate of the personal computing field is decided. These users will make these decisions based on what they see, hear, and touch. They will not make their decisions based on how many chips are in a box or whether interrupts are enabled during DMA disk transfers.

Using these predictions and a little imagination, we can hypothesize what the ultimate personal computer will be like. Figure 6 illustrates my view of such a machine. Four *functional servers* are clustered around a central switching, control, and com-

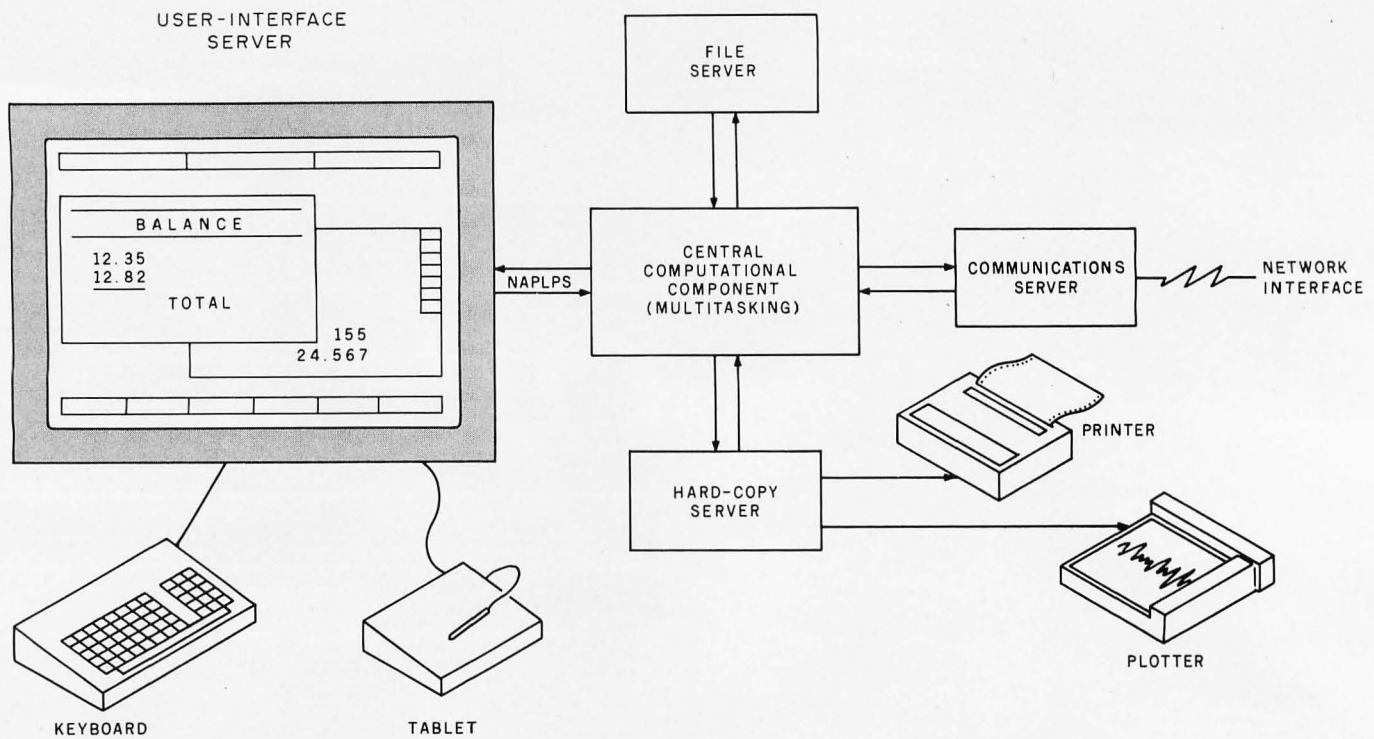


Figure 6: The ultimate distributed personal computer system for the serious user. Each of the components is an independent entity, each with the power of a typical personal computer available today. The diagram illustrates functionality and does not necessarily imply physical partitioning. NAPLPS is the language used between all components, especially between the User Interface and the Central Computational Component. The User Interface may be able to run a variety of local screen-based text and graphics editors without informing the Central Computational Component.

putational unit. These four functional servers are *not* meant to indicate peripherals hanging on a microcomputer. In fact, they may instead be integrated into one unit. To put things into perspective, I imagine that each of the functional servers would have the complexity equal to or greater than that of an IBM Personal Computer.

The Hard-Copy, Communications, and File Servers are straightforward. Ideally, the Hard-Copy Server would support integrated text and graphics in high-resolution black and white. Color hard copy would be desirable, but I could probably live without it. The Communications Server would provide all links to off-site services through modems and local networks. The File Server would provide the typical storage and retrieval functions. Features such as automatic redundancy and backup might be transparent to the user. Database-management capabilities could be built into the File Server, which would be responsible for organizing

the database in the optimum way based on the user's answers to certain queries.

The User-Interface Server would be an extremely high resolution color-graphics display with a variety of user-input devices. All input editing would be handled by this server. The user should be able to enter text and graphics with equal ease. The ability to point to objects on the screen should be available from any of the input devices. And the User-Interface Server should be able to support multiple *windows*, representing various concurrent activities currently in progress.

The Central Computational Component would be a multitasking system with the computational capability similar to most multiuser timesharing systems. A process (or task) would be active for each window in use in the User-Interface Server. Additional processes could be created to perform tasks for the user.

The Central Computational Component would also be responsible for

coordinating all server-to-server interactions. It would also act on the user's behalf if some attention is needed and the user is busy. For example, electronic mail received via the Communications Server would be stored in the File Server without the user becoming involved. The user would be able to retrieve the mail at a later time.

Given this model of an ultimate personal computer, it becomes useful to begin charting an evolutionary path from our current position to this ultimate system. It is obvious in figure 6 that the User Interface must be present in the system before any other component. It is not obvious that the user needs any other component as long as the User Interface has access to a Central Computational Component.

The first step along the evolutionary path is to give the user a User-Interface Server. From that day forward, the user's view of the system begins to form. (The old adage "love at first sight" can be applied to com-

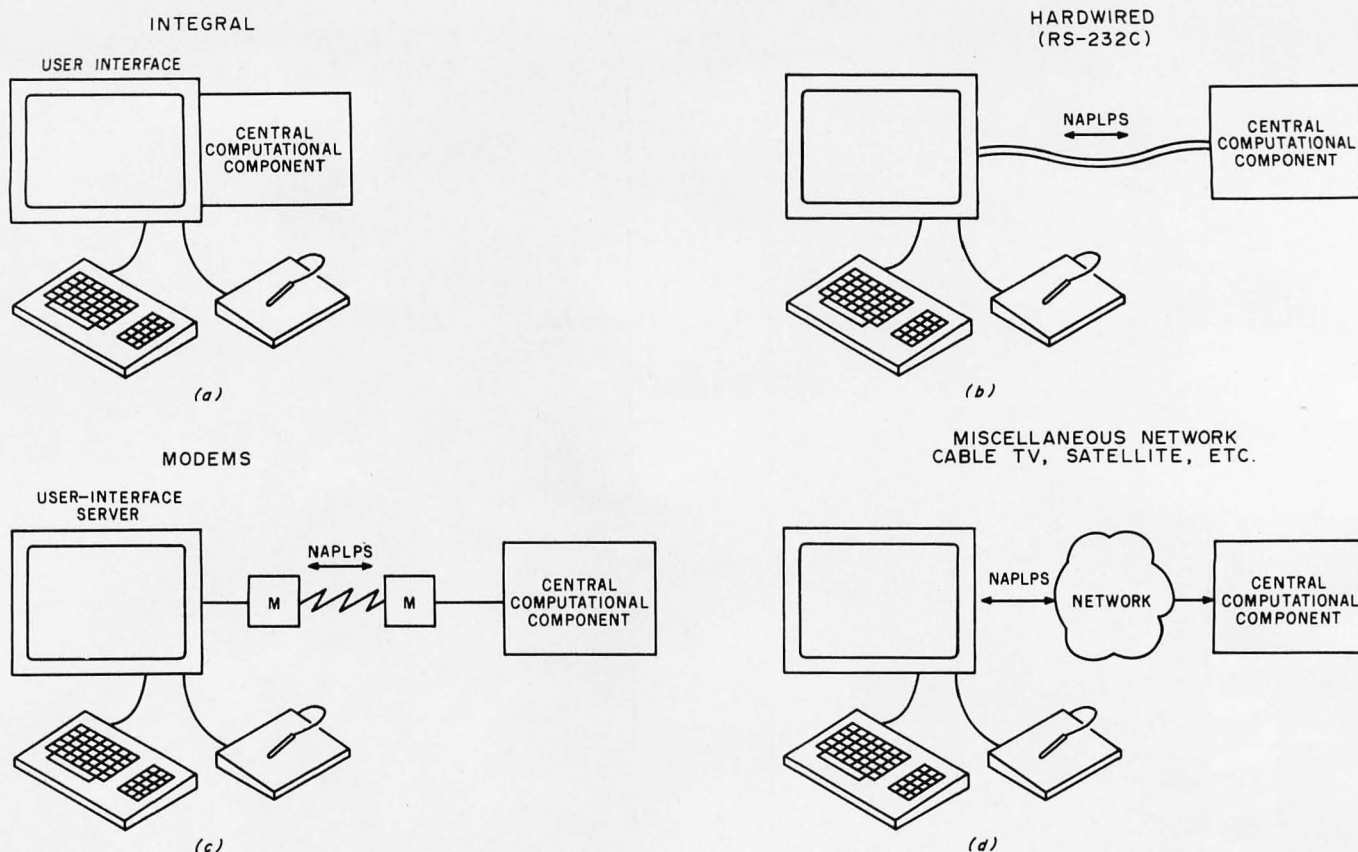


Figure 7: The User-Interface Server can be connected to the Central Computational Component in a variety of ways. A modem or a network arrangement allows a person to start using a User-Interface Server long before the Central Computational Component is needed.

puters as well as people.)

Because of this user view that is established, one is driven to give the user as much capability from day one as is economically feasible, even at the expense of not providing other components in the system. This strategy is compatible with the fact that a user will judge a system primarily from interaction with the User-Interface Server.

Once a user is given a User-Interface Server, another serious situation develops: the user will not want to give it up. This implies that when the user eventually obtains a Central Computational Component, he or she will expect the User-Interface Server to work with the new component.

What this all comes down to is that the user should be given the ultimate text/graphics terminal before any other component. The capabilities of

this terminal should be standardized, and everything but the kitchen sink had better be available, because the capabilities of that terminal will help shape all future services. And the interface to that terminal must be clearly defined *before* any terminals are given out.

This is where NAPLPS becomes involved—as the text/graphics terminal-interface protocol for the User-Interface Server. As shown in figure 7, a NAPLPS terminal can be connected via a variety of mechanisms to a Central Computational Component. From the first day of use, the user sees a certain set of capabilities and begins to get used to the local editing features available in the terminal. If a user is connected to a host via a modem, all computation and information are obtained from the remote host. The spectacular text/graphics displays are there from the

beginning and are not something the user must dream of having.

From the host computer's point of view, the user is now a constant, a known entity with a standard interface. Services can be developed with the assumption that the user will not be a moving target. This stability gives the developers of the host system more incentive to develop more services. More services attract more users, and more users of course mean more money, etc., etc.

The user begins to consume the prepackaged information and services. Most users may not want to become programmers or system administrators; they merely want to accomplish a task and get on with the rest of their lives.

At some point in life, however, a user may become more sophisticated and require a dedicated processor. A Central Computational Component

can be purchased by the user and added to the original User-Interface Server. Many of the services that had been available on the remote host will probably be available on the dedicated processor.

Additional servers could be added to the Central Computational Component as the user desires. One interesting thing to note is that the display capabilities of NAPLPS are such that the Hard-Copy Server can use the same protocol. Also, because the File Server will certainly be able to store NAPLPS, we see that a complete, integrated personal computing system begins to emerge. NAPLPS becomes the common language in the system for information interchange.

The addition of the Communications Server is the mechanism by which a user and personal computer become an entity on a local-area network. Keep in mind that the addition of the Communications Server allows a far more sophisticated system than when the user was *remoted* to a host via an unspecified network. The *remoting* was done to create the link be-

tween the User-Interface Server and the Central Computational Component. The Communications Server was not involved. Figure 7 illustrates this difference.

If I have not lost you completely and you are good at reading between the lines, it should be clear that NAPLPS is a small part of a large, integrated personal computing system. Many of the parts of that system still need to be specified and implemented. Luckily, the part of most concern to users is quite mature and currently available in the marketplace. I predict that in three to four years systems similar to the one I have described will be commonplace in personal computing.

Those of you who are familiar with the various videotex systems that have been proposed may realize that my ultimate personal computer would be fairly compatible with such systems. In most videotex systems, a network of inexpensive home terminals with enhanced graphics capabilities are connected to a host computer. Ideally, the manufacturers of

these terminals will design them so that at some later date the user can add the above-mentioned servers.

As I said at the outset of this article, personal computing has not yet reached the majority of people in the world—but someday it will. If you are currently a personal computer user, you should feel honored to be among the pioneers. As the personal computing field begins to evolve into a mass-market, consumer-oriented, network-based information system, do not be surprised if you begin to feel like the odd man out. Also, do not be surprised if one day your neighbors invite you over to see their new computer that costs one fourth to one tenth the price of yours and has access to thousands of services via networking. And do not get upset when you find out that your neighbors cannot tell you which processor their computer uses or what transmission rates their modem supports. Instead, just feel good about the fact that you know a little bit about the language their computer uses to talk to the rest of the world. ■

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